

ProtoHedge Beyond Simulation: Empirical Validation on Historical Option Data

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Abstract—Deep Hedging can learn effective risk-sensitive trading policies, but its individual decisions are difficult to interpret. ProtoHedge addresses this limitation by expressing each hedge as a similarity-weighted combination of representative market states, providing an exact example-based decomposition while retaining end-to-end learning. Existing evidence for the method is largely simulation based, leaving its behavior in historical markets unresolved. We conduct a controlled empirical evaluation of ProtoHedge against Deep Hedging and a transparent delta-based benchmark across 34,395 historical option episodes from ten U.S. underlyings and both standard and path-dependent option liabilities. Under matched data, information, objectives, and evaluation procedures, ProtoHedge remains competitive with Deep Hedging for European options and shows encouraging tail-risk performance for path-dependent options, while comparing favorably with the delta benchmark. These findings extend the evidence for prototype-based hedging beyond simulation and suggest that interpretable policy structure can retain competitive empirical performance on historical markets.

Index Terms—deep hedging, interpretable machine learning, option hedging, ProtoHedge, conditional value-at-risk, sequential decision making

I. INTRODUCTION

Hedging is a sequential decision problem under uncertainty. At each trading date, a hedger must act from incomplete market information, absorb transaction costs, respect position limits, and control a terminal loss distribution. Deep Hedging (DH) addresses this problem by optimizing a neural policy directly against a monetary utility objective [1]. This formulation can incorporate market frictions and complex state variables without requiring a closed-form hedge. At each rebalancing date, the network maps observable market conditions and current inventory to a trade; training over complete paths balances terminal risk against cumulative transaction costs. Its flexibility comes at a cost: the reason for an individual neural-network action can be difficult to inspect.

ProtoHedge (PH) was proposed as an intrinsically interpretable alternative [2]. PH retains DH’s objective and sequential setup but replaces its direct action map with a fixed library of representative market states. Each trade is a similarity-weighted combination of actions attached to those prototypes, so the examples and weights that generate a decision can be retrieved exactly. The original PH experiments found near parity with DH in Black–Scholes and stochastic-volatility simulators. The practical question is whether this performance–interpretability tradeoff survives historical data, where regimes change, option contracts expire, quotes are noisy, and nearby rolling trajectories are dependent.

Prior work has evaluated neural and reinforcement-learning hedges under richer simulators [3]–[5] and historical index option data [6], [7]. Our study asks a more specific question: when DH and PH receive the same historical state, trade the same instruments, and optimize the same tail-sensitive objective, does the PH prototype bottleneck retain competitive out-of-sample performance? We evaluate PH as a historical hedging model rather than as a test of any particular market simulator.

Historical validation requires choices that are largely absent with independent simulated paths. A listed option must remain the same contract throughout an episode; windows must be formed after chronological splitting; the policy must observe enough state to hedge a path-dependent payoff; and overlapping test episodes require dependence-aware inference. Model and checkpoint selection must also use validation data only. These choices determine what the comparison means and are therefore part of the experimental contribution.

We make three contributions. First, we construct a contract-consistent historical panel for ten liquid U.S. underlyings, with fixed 20-decision horizons and a held-out 2022–2024 test period. Second, we compare DH with a validation-selected PH family for both European and arithmetic-average Asian calls under the original OCE CVaR@50% utility. Third, we report paired circular-block intervals, use validation-only checkpoint selection, and compare PH directly with a transparent Spot-Delta rule. Under this protocol, PH is near parity with DH for European calls, while the validation-tail-selected PH policy has positive panel-level test utility relative to DH for Asian calls.

The experiments address three empirical questions. *Q1*: Does replacing DH’s direct action head with the PH prototype bottleneck cause a detectable loss of OCE utility for terminal-payoff European calls? The original simulation evidence suggests that any loss should be small. *Q2*: Does that conclusion extend to arithmetic-average Asian liabilities, whose relevant state includes the realized path? *Q3*: Are conclusions robust to a transparent market-delta benchmark and to temporal dependence among overlapping episodes? We evaluate these questions on chronologically held-out paths and use paired intervals that account for overlapping episodes, allowing us to distinguish panel-wide evidence from rankings based only on average scores. The European experiment transfers the original PH–DH comparison to historical data, while the Asian experiment extends it to a liability whose payoff depends on the realized path. Upon acceptance, we will release the source code, locked configurations, data-construction scripts, and

redistributable result artifacts needed to reproduce the study. The underlying OptionMetrics and CRSP records cannot be redistributed because they are proprietary and remain subject to their providers' licenses.

II. RELATED WORK

Classical delta hedging derives trades from model sensitivities [8]. It is a local, model-based rule: the position is determined by the option value's instantaneous response to the underlying, rather than by direct optimization of the full distribution of terminal hedging outcomes. Transaction costs and discrete rebalancing motivate cost-sensitive policies [9], [10]. DH instead learns the entire sequence of trades under a terminal risk objective, allowing frictions, position constraints, and nonlinear risk preferences to enter the training problem directly [1]. Reinforcement-learning and historical studies further demonstrate that neural hedging policies can be trained and evaluated on observed option data [4], [6]; broader evidence is reviewed in [11].

PH brings prototype-based, case-based reasoning [12], [13] to DH [2]. Prototype models are intrinsically interpretable in the sense that their output is constructed from representative cases, rather than explained only after a separate opaque model has made its prediction [14]. PH applies this idea at every rebalancing date by combining actions attached to representative market states according to their similarity to the current state. The resulting decomposition exposes the retrieved states, similarity weights, and prototype actions behind each trade. The original study reports PH competitive with DH in controlled simulations, suggesting that this interpretable action bottleneck need not impose a large loss in hedging utility.

The literature therefore leaves a specific comparison unresolved: historical studies validate DH, and simulation studies validate PH, but neither tests PH against DH on historical paths. We match episodes, information, instruments, the objective, and validation rules to test whether PH's interpretability-performance tradeoff survives historical markets.

III. METHODS

A. Hedging Objective and Policies

For hedge prices P_t , post-trade inventory h_t , trade increment a_t , and terminal short-liability cash flow $H \leq 0$, define

$$O^\pi = H + \sum_{t=0}^{T-1} h_t^\top (P_{t+1} - P_t) - \sum_{t=0}^{T-1} \kappa_t^\top |a_t|. \quad (1)$$

We call O^π the *hedging offset* because it excludes the unobserved premium of the synthetic liability. Higher values mean that a strategy offsets more of the common terminal liability after transaction costs. Because all policies face the same liability paths, paired differences isolate hedging performance under this accounting convention.

Both neural policies use the same optimized certainty equivalent (OCE):

$$\mathcal{U}_\lambda(O) = \sup_{y \in \mathbb{R}} \mathbb{E}[(1 + \lambda) \min(O + y, 0) - y], \quad (2)$$

with $\lambda = 1$, corresponding to the lower 50% tail used by the original implementation [15], [16]. We evaluate each trained model with its frozen learned threshold y and call the resulting mean pathwise contribution *OCE CVaR@50% utility*. Higher values are better.

DH implements the trade a_t with a feed-forward neural policy. PH instead standardizes the current state x_t , maps it and fixed prototype states p_k through a shared learned projection f_ϕ , and sets

$$\alpha_{t,k} = \frac{e^{-\|f_\phi(x_t) - f_\phi(p_k)\|^2}}{\sum_j e^{-\|f_\phi(x_t) - f_\phi(p_j)\|^2}}, \quad a_t = \sum_{k=1}^K \alpha_{t,k} b_k, \quad (3)$$

where b_k is the learned bounded action of prototype k . The weights and attached actions are the policy computation itself rather than a post-hoc approximation. We therefore use *inspectable* in the structural sense that each proposed trade has an exact, model-native decomposition. Any subsequent position-limit clipping is a shared deterministic feasibility step. The empirical study tests whether a policy with this structure can remain competitive in utility; whether the resulting explanations improve human decisions requires a separate study.

B. Historical Panel

We combine OptionMetrics IvyDB US listed-option records with CRSP underlying prices [17], [18]. The underlyings are AAPL, IWM, NVDA, QQQ, SPY, TLT, XLE, XLF, XLK, and XLV. They include two single stocks, broad and sector equity exchange-traded funds, and a Treasury exchange-traded fund over 2005–2024. This mix tests the same hedging protocol across idiosyncratic equities, diversified portfolios, sector exposures, and an interest-rate-sensitive asset rather than within a single market segment.

The quote screens ensure that both the policy state and friction proxy are defined from usable observations. Retained records are American-style calls with finite implied volatility and vendor Greeks, positive bids and midquotes, and non-crossed offers. At each candidate start date, initial open interest must be positive, relative spread cannot exceed one, and absolute strike moneyness cannot exceed 10%. This removes internally inconsistent or extremely illiquid quotes before any model is trained.

Among surviving calls with 20–40 calendar days to expiration, contracts are ranked by strike distance from spot, distance from a 30-day target maturity, relative spread, open interest, and volume. The ranking uses no subsequent hedging outcome. Once selected, the contract is held fixed: all 21 observations must share its option identifier, strike, and expiration, and expiration must follow the episode. Gaps longer than four calendar days begin a new block. Requiring a complete fixed-contract path sacrifices sample size but prevents an implicit daily roll between unrelated options, so each episode represents one well-defined sequential hedging problem.

To preserve temporal order, we split the calendar before constructing rolling windows. Training ends in 2018, validation covers 2019–2021, and testing covers 2022–2024. A 20-observation embargo at both post-training boundaries ensures

TABLE I
HISTORICAL PANEL AND LOCKED PROTOCOL.

Component	Specification
Underlyings	AAPL, IWM, NVDA, QQQ, SPY, TLT, XLE, XLF, XLK, XLV
Periods	Train through 2018; validation 2019–2021; test 2022–2024; 20-observation embargo
Episodes	20,537 train; 6,881 validation; 6,977 test
Episode unit	One option ID; 21 marks; 20 decisions
Liabilities	ATM European and arithmetic-average Asian calls
Hedge assets	Underlying and one fixed listed call
Neural runs	Three seeds; 800 epochs; common OCE objective
Inference	2,000 paired circular-block draws; block length 20

that no episode crosses a split or shares its horizon across that boundary. Thus, model selection precedes the test period and each test episode represents a later market path. The result is 20,537 training, 6,881 validation, and 6,977 test episodes, or 34,395 total (Table I). Episodes may still overlap within a period; the inference procedure accounts for that dependence.

Each episode defines a synthetic at-the-money short call struck at initial spot $K_i = S_{i,0}$. The European payoff is $H_i = -\max(S_{i,T} - K_i, 0)$. The Asian payoff replaces terminal spot with the arithmetic mean of the 20 end-of-interval fixings [19]. This pair contrasts a terminal-state liability with one whose value depends on the realized path. The hedge instruments are the underlying and the fixed listed call; trade increments and cumulative positions are constrained to $[-1, 1]$ for both.

The market record contains $(S_t, C_t, \Delta_t, \nu_t, \sigma_t^{\text{impl}})$: underlying spot, fixed-option midpoint, vendor delta, vendor vega, and implied volatility. The policy state is narrower: the networks receive current spot and option prices, the two inventory positions, and time remaining; Asian policies additionally receive running-average moneyness. Spot, option price, and vega are scaled pathwise by initial spot. Variables outside the policy state still serve defined protocol roles: vendor delta constructs Spot-Delta, delta, vega, and midpoint construct option costs, and implied volatility supports screening. The running average uses only fixings observed by the decision time. DH, every PH candidate, and prototype extraction therefore receive the same task-appropriate inputs without future path information.

Accounting is also common. The trade at time t first updates cumulative inventory, and that post-trade inventory equals the price change from t to $t + 1$ in (1). Applying only the latest trade increment to that gain would incorrectly discard positions established earlier in the episode. The short liability is settled only at the horizon.

Costs follow the original study’s deterministic friction proxy. A stock trade costs $0.0002|S_t|$ per unit; an option trade costs $0.02|\nu_t| + 0.0002|\Delta_t| + 0.0005|C_t|$, using vendor vega ν_t , delta Δ_t , and midpoint C_t . These stylized costs provide a common friction model for the controlled policy comparison; every policy receives identical prices and costs.

C. Models and Baselines

The transparent baseline, Spot-Delta, holds the selected listed call’s observed delta in the underlying and no option

inventory. Because the signal is the vendor delta of the listed hedge option, Spot-Delta is a market-based reference rather than an analytical hedge for the synthetic liability. DH uses the original paper-style feed-forward architecture with width 20, depth 3, and Softplus activations.

PH uses $K \in \{10, 25, 50, 100\}$ prototypes obtained by K -means [20] from either DH or Spot-Delta training trajectories, with unweighted or time-weighted similarity. Each centroid is replaced by its nearest observed training state, preserving a concrete historical example. Prototype anchors remain fixed while the projection and prototype actions are learned. The two sources, two weighting rules, and four values of K yield 16 PH configurations. DH-sourced, unweighted candidates are closest to the original PH construction; Spot-Delta sourcing and time weighting extend it.

Two additional rules provide descriptive context in the archived outputs. An unhedged strategy never trades. A no-trade-band delta rule retains its current spot position while it lies inside a validation-tuned band around observed delta and otherwise moves to the nearest edge. The primary comparisons use DH and Spot-Delta as the neural and transparent benchmarks, respectively.

D. Model and Checkpoint Selection

Every neural policy uses three seeds, an 800-epoch budget, Adam with learning rate 10^{-3} , and the objective in (2). Training also includes mean absolute-action and inventory penalties of 0.001 and 0.002. These regularizers stabilize optimization but are excluded from the test utility in Table II.

Within each ticker and liability, validation metrics are averaged across seeds. *Validation-Mean* selects the PH configuration with the highest mean hedging offset, whereas *Validation-Tail* selects the configuration with the highest empirical CVaR@5%. Candidates whose seed-averaged validation bound occupancy exceeds 0.5 are screened out. Ties are resolved deterministically by the alternate validation metric, lower occupancy, smaller K , and model name. The resulting configuration is frozen before test evaluation.

PH searches 16 configurations per seed, whereas DH uses one fixed architecture. Accordingly, our estimand is the performance of a validation-selected expanded PH family against the original paper-style DH benchmark, not against an equally tuned DH family. This distinction guards against overinterpreting family-level model selection [21].

The initial checkpoint rule ranked saved epochs using validation OCE together with small action, inventory, and bound-occupancy penalties. An audit found that this rule selected the initialized DH policy in 12 of 60 fits. Although initialization is an eligible checkpoint, the pattern showed that diagnostic penalties could override improvement in the primary utility and made the European comparison sensitive to the checkpoint criterion.

We therefore report a target-aligned sensitivity analysis. For the affected DH fits and their matched selected PH candidates, the replacement checkpoint is the one that maximizes validation OCE in the existing training history. This requires no

retraining: selection uses stored validation trajectories alone, after which checkpoints are frozen for test evaluation. The original penalized-score results remain archived to document the sensitivity.

E. Statistical Inference

For PH–DH comparisons, the primary statistic is the paired difference in frozen-threshold OCE utility. For PH–Spot-Delta comparisons, we report mean hedging offset and empirical CVaR@50%. For each episode, neural-policy outcomes are first averaged across seeds. We then resample paired episodes within each ticker using 2,000 circular-block draws of length 20 and average the ten ticker effects with equal weight [22], [23]. Positive differences favor PH; reported 95% intervals are bootstrap percentile intervals without multiplicity adjustment.

The resampling design targets an equal-asset effect. Seed averaging prevents a historical episode from being counted three times, paired resampling compares policies on the same paths, and equal ticker weights prevent larger panels from dominating. The block length equals the full 20-decision horizon, preserving dependence induced by strongly overlapping neighboring episodes.

F. Empirical Estimand

The primary estimand is the equal-asset, held-out PH–DH OCE effect produced by each frozen validation rule, rather than the best test result available in the grid. It therefore evaluates a deployable model-selection procedure instead of rewarding a larger candidate family for an ex post test maximum. “Tail” names the validation criterion; both selected policies are tested on the same OCE utility.

OCE CVaR@50% is primary because it is optimized by both architectures and is the utility used to motivate PH. Mean hedging offset and empirical CVaR@50% against Spot-Delta provide complementary distributional summaries. RMSE, downside deviation, trading cost, action magnitude, and bound occupancy are archived as diagnostics rather than treated as additional headline tests.

PH is competitive when it retains its exact prototype decomposition without a detectable loss under the primary utility. A positive interval supplies evidence of improvement; an interval crossing zero supports only the weaker conclusion that this study does not detect a difference.

IV. RESULTS

A. Panel-Level Comparisons

Table II contains the primary ten-ticker results. Against DH, the mean-selected European policy produces a PH difference of +0.001009, and the tail-selected European policy produces +0.000514. Both intervals in Table II include zero. The data therefore support near parity, not European PH superiority.

For Asian calls, Validation-Mean is positive but uncertain at +0.001859. Validation-Tail is +0.004015, and its interval excludes zero. This is the only primary PH–DH interval separated from zero. The OCE-only checkpoint analysis reduces the initial European estimates of +0.008841 and +0.006693

by approximately 89% and 92%, while changing the Asian tail estimate by only 3%. Eleven checkpoint replacements enter each European comparison but only one enters each Asian comparison, explaining why the European result is more sensitive to the selection rule.

The Spot-Delta comparisons provide a transparent second benchmark. European Validation-Tail PH improves mean hedging offset by 0.005488, but its empirical CVaR@50% difference is near zero. Asian Validation-Tail PH improves both mean hedging offset (+0.002795) and empirical CVaR@50% (+0.005830), with positive intervals. Thus the Asian result is not solely an artifact of the DH benchmark, although the OCE and empirical CVaR statistics should not be treated as numerically interchangeable.

B. Cross-Asset Evidence

Fig. 1 reports Validation-Tail PH–DH OCE differences for every ticker. European effects are heterogeneous: four of ten estimates are positive, only TLT has an interval wholly above zero, and no uniform advantage is visible. Asian estimates are positive for seven of ten tickers; AAPL, NVDA, TLT, and XLF have positive individual intervals, while no ticker interval lies wholly below zero. Equal-ticker aggregation prevents large underlyings or those with more windows from dominating the panel result.

The cross-asset pattern matters for interpretation. The positive Asian panel effect is not produced by one asset, but neither is it universal. Its strongest estimates occur for AAPL, NVDA, and TLT, while SPY is near zero and XLE and XLK are negative with wide intervals. This heterogeneity supports a panel-level path-dependent tail finding and motivates future regime analysis.

Validation-Mean does not change the panel conclusion. Seven of ten point estimates are positive for each liability, but both equal-ticker intervals cross zero. The supported Asian improvement is therefore specific to the Validation-Tail selection rule rather than a general mean-performance advantage.

C. Checkpoint Diagnostic

Bound occupancy helps explain the checkpoint sensitivity. DH occupies an inventory bound in approximately 39% of test path–time–instrument observations for both liabilities, whereas selected PH policies average roughly 4–5% for European and 2% for Asian calls. Because occupancy is a diagnostic rather than the test objective, including it in the initial checkpoint score disproportionately favored lower-occupancy policies. OCE-only selection aligns checkpoint choice with the reported utility.

V. DISCUSSION

The European result supports PH’s original performance–interpretability premise on historical data. Replacing the final black-box action map with a prototype mixture does not produce a statistically detectable utility loss across the ten-ticker panel. Near parity is meaningful here because PH retains

TABLE II

EQUAL-TICKER HISTORICAL EFFECTS ON THE HELD-OUT 2022–2024 TEST PERIOD. DIFFERENCES ARE PH MINUS BENCHMARK; HIGHER IS BETTER. PH–DH ROWS USE FROZEN-THRESHOLD OCE CVAR@50% UTILITY AFTER THE TARGETED CHECKPOINT SENSITIVITY. PH–SPOT-DELTA ROWS USE THE VALIDATION-TAIL PH POLICY.

Benchmark	Liability	Selection	Metric	Difference	95% block interval
DH	European	Validation-Mean	OCE utility	+0.001009	[−0.002777, +0.005364]
DH	European	Validation-Tail	OCE utility	+0.000514	[−0.003418, +0.004734]
DH	Asian	Validation-Mean	OCE utility	+0.001859	[−0.000309, +0.003863]
DH	Asian	Validation-Tail	OCE utility	+0.004015	[+0.001925, +0.006038]
Spot-Delta	European	Validation-Tail	Mean hedging offset	+0.005488	[+0.004086, +0.006858]
Spot-Delta	European	Validation-Tail	Empirical CVaR@50%	+0.000163	[−0.001925, +0.002134]
Spot-Delta	Asian	Validation-Tail	Mean hedging offset	+0.002795	[+0.000271, +0.005291]
Spot-Delta	Asian	Validation-Tail	Empirical CVaR@50%	+0.005830	[+0.002925, +0.008393]

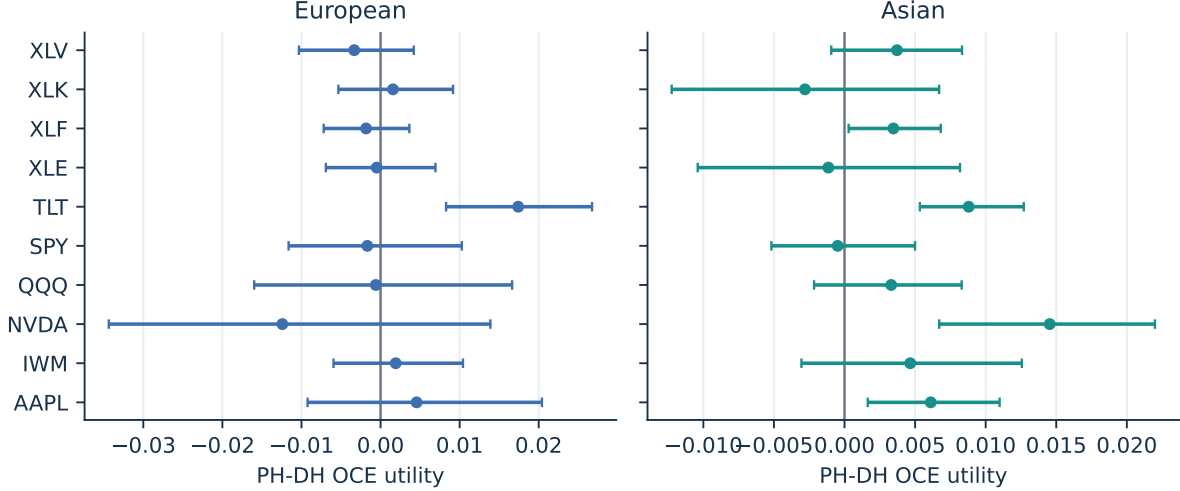


Fig. 1. Validation-Tail PH minus DH frozen-threshold OCE CVaR@50% utility by ticker. Points are estimates and bars are paired 95% circular-block intervals. Positive values favor PH.

a direct, model-native decomposition of each proposed trade into similarities and prototype actions.

The Asian result raises the study’s main forward-looking hypothesis: routing actions through a finite reference set may regularize a policy when the liability state depends on the path. Both models receive the same running-average money-ness state, but the prototype bottleneck constrains how PH turns that information into trades. The present experiment cannot separate this mechanism from PH’s larger configuration search. It nevertheless identifies path-dependent tail hedging as a promising setting for a matched, independently confirmed comparison.

The benchmarks answer complementary questions. DH largely isolates the effect of the prototype action bottleneck, whereas Spot-Delta anchors the evaluation to a transparent rule independent of neural training. The Asian advantage appears against both; the European evidence supports competitiveness but not a tail improvement. This agreement strengthens the Asian signal without removing the need for a matched-search confirmation.

A. Limitations and Next Steps

Model selection. PH receives a larger development budget than the fixed DH benchmark, and 25 of 40 selected configurations use an added prototype source or similarity weighting. The evidence therefore applies to a validation-selected PH family rather than one portable configuration. The checkpoint analysis reuses stored training histories. A fully matched study should equalize search budgets, use OCE checkpoint selection from the outset, and lock one PH design across assets.

Market realism. Daily midquotes and stylized costs provide a controlled comparison but not an executable slippage model. The liability is synthetic and at the money, while the hedge option is a selected listed contract. Because the unobserved liability premium is omitted, absolute hedging-offset levels are comparative rather than profitability measures. Further tests should add maturity and money-ness buckets and executable spread models.

Generalization and inference. Chronological testing measures transfer to later regimes of the same ten underlyings, not to unseen assets. The evidence is also limited to liquid U.S. names, one horizon, and call liabilities. Block resampling preserves within-ticker temporal dependence but not syn-

chronous cross-ticker shocks, and the four PH–DH intervals are not multiplicity adjusted. Independent confirmation should therefore include new assets and periods and a panel-aware multiple-testing design.

VI. REPRODUCIBILITY

Our reproducibility archive preserves the locked historical protocol, SHA-256 manifests, outputs for all ticker–liability tasks and seeds, candidate and checkpoint identities, pathwise paired outcomes, bootstrap results, and the asset-generation script. Tables and figures are generated from archived CSV files rather than transcribed manually. The original run remains immutable, with the checkpoint sensitivity stored as a separate analysis. Upon acceptance, we will release the code, configurations, data-construction scripts, and redistributable artifacts; the underlying OptionMetrics and CRSP records remain subject to their providers’ licenses.

VII. CONCLUSION

This study extends ProtoHedge from simulation to historical option markets across ten U.S. underlyings and two liability structures. With validation-only selection and paired temporal-block inference, PH remains competitive with DH for European calls and preserves an exact prototype decomposition of each proposed trade. For arithmetic-average Asian calls, the tail-selected PH family yields a positive panel-level OCE utility difference against DH. It also improves mean and empirical lower-tail outcomes against Spot-Delta.

These findings provide historical support for prototype-based hedging and identify path-dependent liabilities as the most promising setting for further study. A matched-search, fixed-configuration experiment should next test whether the prototype bottleneck itself supplies useful regularization across assets and market regimes. More broadly, the results suggest that interpretable policy structure can remain compatible with risk-sensitive sequential decision making in observed markets.

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